

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Why would something appear blue?' is 'it reflects the same wavelength as the sea'.
2. True or False: The correct answer to 'In which object does chemical energy most likely transform into electrical energy?' is 'a television'.
3. True or False: The correct answer to 'Populations living in one place form a' is 'system'.
4. True or False: The correct answer to 'In American football how many points is a touchdown worth?' is 'six'.
5. True or False: The correct answer to 'The chemical reaction of ethane and oxygen produces carbon dioxide and water. Which is the best form to identify reactants and products...' is 'an equation'.
6. True or False: The correct answer to 'Which of the following is a mineral?' is 'diamond'.
7. True or False: The correct answer to 'Simple tandem repeat polymorphisms in humans are most useful for' is 'solving criminal and paternity cases'.
8. True or False: The correct answer to 'A magnetic force can act on an electron even when it' is 'Both of these'.
9. True or False: The correct answer to 'Which most directly requires sunlight to grow?' is 'a bird'.
10. True or False: The correct answer to 'Which behavior is an example of an animal adapting to a change in the environment?' is 'a frog eating an insect'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to check whether the given month number contains 28 days or not.
2. Write a function to count the same pair in three given lists.

End of Paper